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Palaeomagnetism of the British Tertiary igneous province: Rhum and Canna

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Summary. Palaeomagnetic measurements have been made on samples of lavas, dykes and plutonic rocks from Rhum, and of lavas and dykes from Canna. All the samples are reversely magnetized, except for three dykes cutting the Rhum lavas and two dykes cutting the Canna lavas.

The magnetizations are believed to be primary, or to have been acquired only a short time after the units were formed.

The simplest interpretation of the polarities is that the majority of the rocks were formed in a single reversed interval, with the normally magnetized dykes being intruded in the following normal interval. However, if the time during which the Rhum centre was eroded, prior to the extrusion of the lavas, was long, additional inversions of the Earth's magnetic field could have escaped detection. In this case the polarity sequence is extended from R–N to R...N ...R–N, with the final R–N corresponding to the lavas and the normal dykes cutting them.

Igneous activity began about 59 Ma ago, contemporary with activity elsewhere in the British Tertiary igneous province.

Introduction

The Small Isles of Inverness-shire are located at 57° N, 353.5° E, between Mull to the south and Skye to the north. They are largely composed of Tertiary igneous rocks.

The Tertiary geology of Rhum has been described by Dunham & Emeleus (1967) and Emeleus & Forster (1979); the sequence of events proposed by them is incorporated in Table 1. An initial phase of intrusive activity, during which gabbro, explosion breccia, felsites and granophyres were emplaced, was followed by ring faulting. Later, ultrabasic rocks were intruded. Dykes were injected at various times during the development of the complex, although mainly before the ring faulting. The lavas can be divided into four groups, olivine basalts, hawaiites, basaltic andesites (early mugearites, Fig. 1) and icelandites (late mugearites, Fig. 1). These lavas are cut by a number of dykes, but must themselves be considerably younger than the plutonic complex as they were extruded into river valleys eroded

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Table 1. The Rhum Tertiary igneous sequence (modified from Dunham & Emeleus 1967; Emeleus & Forster 1979).

	Unit	Code no.	Polarity
Youngest			
	Dykes (cutting lavas)	402D, 405D, 417 D, 418D	R + N
Lavas	Icelandite lavas	404	R
	Basaltic andesite lavas	406–412	R
	Hawaiite lavas (Orval)	413–416	R
	Olivine basalt lavas	400–403	R
	Erosion		
Central intrusive complex	Fine grained olivine gabbro	300	R
	Marginal gabbro	301	R
	Layered ultrabasic	322–325	R
	Ultrabasic tongues	308, 309, 330	R
	Main Ring Fault		
	Dykes (sampled along coast)	33–248*	R
	Granophyres	307	R
	Felsites	302, 321	R
	Tuffisites		
	Explosion breccia	310–320	R
	Early gabbro		
Oldest			

* Numbering follows Speight (1972).

in the unroofed complex from a source possibly in or near Canna (Emeleus 1973). This is in contrast both to Mull and Skye where the extrusion of lavas preceded the development of the centres. Although it has been suggested (Brown 1956; Stewart 1970), that an earlier group of lavas, similar to the porphyritic Central-type basalts of Mull, may have been extruded as the layers of the ultrabasic mass built up below ground, and subsequently eroded, there is little evidence to support this view.

The geology of Canna is much simpler than that of Rhum. Five successive groups of basalt/hawaiite flows have been identified by Allwright (private communication). Conglomerates interbedded between the flows of the lava pile contain pebbles of gabbro and granophyre whose most likely source is Rhum (Allwright, private communication), so that the lavas are a relatively late feature of the activity of the Small Isles. Sixteen dykes cutting the Canna lavas have been mapped by Speight (1972), who also mapped the coastal dykes of Rhum.

Speight's work in Skye and the Small Isles identifies several major dyke swarms, each of which trends approximately north-west–south-east, but which have distinct axes of dilation; for example, that of Rhum is separate from both that of Skye and that of Muck. There is also a number of subsidiary swarms with trends towards north and east of north. Trends both west and east of north are found amongst the dykes cutting Rhum and Canna. One noticeable feature is the small number of dykes exposed in Canna (5 km^{-1} on Sanday) compared with the facing coast of Rhum (60 km^{-1}) only 10 km away to the south-east (Fig. 1).

A small amount of palaeomagnetic work has been reported for Rhum only. Khan (1960) studied the layered ultrabasic to test the suitability of the rocks for palaeomagnetic work and to compare the results with the Skye gabbros. Oldershaw (1967) worked on the Western Granophyre and some layered ultrabasic bodies, primarily to determine whether the

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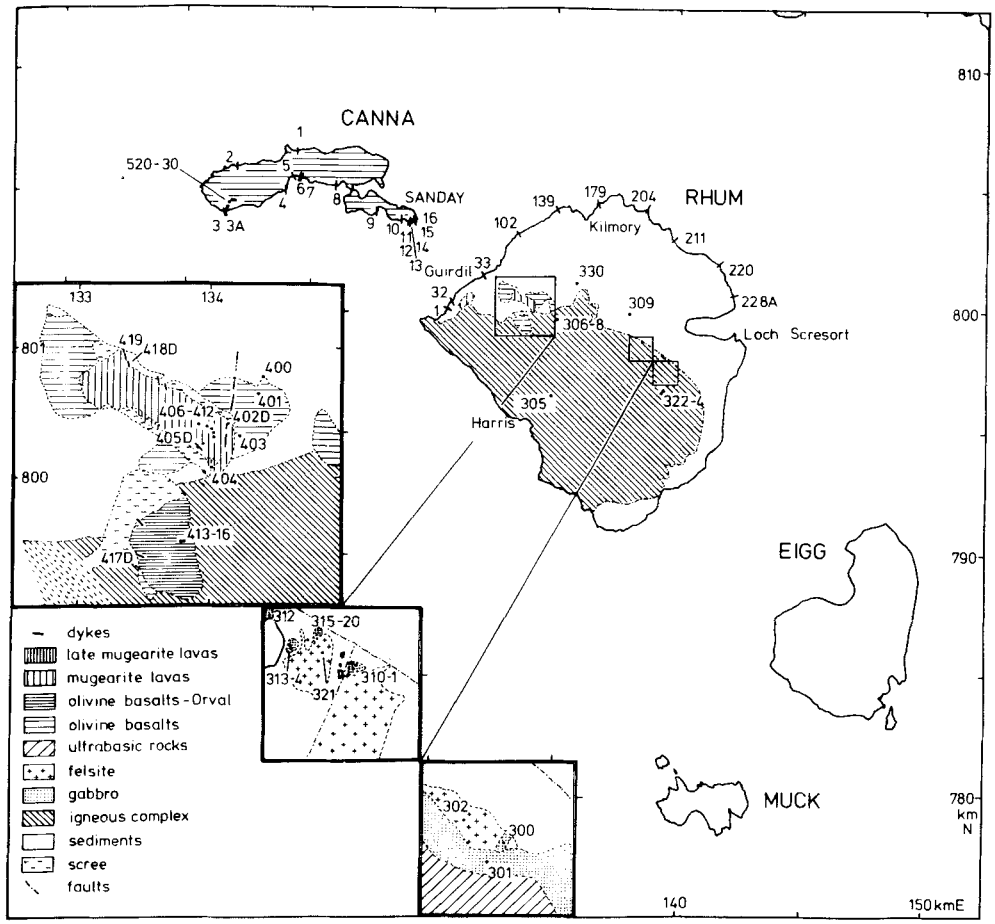


Figure 1. Map of the Small Isles showing the sample locations. Only representative dykes from the Rhum coastal swarm are shown.

observed geological dips were a primary or secondary feature. The principal objective of the work described in this paper was to examine the palaeomagnetism of the lavas and dykes both of Rhum and Canna, together with a representative sequence through the Central complex of Rhum, and to build up a magnetic stratigraphy. In addition, we explored the possibility that the layered intrusion recorded variations of the geomagnetic field and examined two breccias to see whether magnetization was acquired before or after formation.

In Rhum the lavas sampled constitute a sequence through the four lava types, and our aim was to sample every third dyke of the 248 Speight (1972) has mapped between Guirdil Bay and Loch Scresort. However, because of difficulties of access and drilling, the coverage is not even and only 61 dykes and one sill were sampled (A03T033-248), the numbering being the same as that of Speight.

In Canna the single profile of lavas sampled is made up mainly of flows of the middle three groups identified by Allwright (private communication) and does not include the oldest flow. All but two of the dykes were sampled.

Cores (25 mm diameter) were drilled at each sampling site and oriented by Sun-sighting. The locations of the sampling sites are shown in Fig. 1, and the rock types, numbers of units and cores measured are given in Table 2.

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Table 2. Palaeomagnetic results. Normal polar colatitude, $0° < \theta < 40°$. Intermediate, $40° < \theta < 140°$. Reversed, $140° < \theta < 180°$ (Wilson *et al.* 1972b). Curie points on cooling in brackets.

Unit	Code	Samples	D (° ETN)	I (° down)	α_{95}	Polarity	T_C (°C)
Geocentric axial dipole			0	± 72°			
Rhum centre							
Late fine grained gabbro	A03T300	5	192.7	−64.2	2.0	R	590
Marginal gabbro	301	4	164.6	−46.8	2.8	R	580
Felsite	302	4	178.0	−54.7	4.0	R	600
Brecciated cumulate	305	12	180.2	−44.8	6.5	R	595
Dyke cutting 307	306	4	175.7	−61.2	1.2	R	580
Granophyre	307	3	182.2	−53.8	1.6	R	600
Ultrabasic	308	4	169.4	−57.6	2.0	R	~600
Ultrabasic	309	4	177.6	−57.3	1.6	R	600
Explosion breccia	310–320	11	185.7	−66.8	5.8	R	585–600
Felsite	321	2	177.5	−61.6	2.8	R	590
Hallival ultrabasic	322–325	4	180.6	−60.5	4.1	R	550–600
Ultrabasic	330	4	160.6	−65.6	6.3	R	590
(Ultrabasic Khan (1960)			181.0	−55.1	–	R)	
(Granophyre and ultrabasic Oldershaw (1967)			186.5	−62.6	3.0	R)	
(Overall mean)	300–330		176.7	−58.17	4.4		
Rhum lavas							
Olivine basalt (Fionchra)	A03T400		Disparate directions				280 (445)
	401	6	197.9	−55.5	12.0	R	285 (440)
	403	4	127.0	−76.0	1.4	R	275 (430)
Icelandite	404	4	192.2	−47.0	1.6	R	~250 (450)
Basaltic–Andesite (Fionchra)	406	4	147.5	−76.1	2.3	R	245 (490)
	407	5	152.3	−78.5	5.4	R	250 (505)
	408	4	174.6	−63.7	6.7	R	~240 (460)
	409	4	187.1	−62.9	2.2	R	~300 (490)
	410	3	184.1	−51.8	3.8	R	~250 (450)
	411	3	154.3	−69.4	5.2	R	250 (500)
	412	3	019.6	−72.1	4.9	I	280 (500)
			(lightning suspected)				
(mean 406–411			171.9	−67.8	9.7	R)	
Orval basalts (Hawaiite)	413	7	124.8	−69.5	1.5	R	400 (410)
	414	6	114.7	−56.5	5.3	I	310 (355)
	415	6	115.5	−67.6	2.9	R	~340 (450)
	416	4	133.1	−59.9	7.5	R	255 (350)
(mean 413, 5, 6			125.3	−65.9	9.6	R)	
(Overall mean)	401–416		161.1	−68.27	3.06		
(excluding 412, 414)							
Dykes cutting lavas	402D	4	170.6	−72.5	3.0	R	300 (400)
	405D	4	028.5	69.1	1.6	N	230 (230)
	417D	4	315.7	76.9	5.5	N	250 (420)
	418D	4	342.2	67.4	2.6	N	
Rhum coastal dykes							
	A03T033	4	No mean disparate directions				525
	039	4	167.2	−66.6	2.2	R	580
	042	4	133.3	−42.3	10.3	I	570
	045	4	138.0	−63.7	3.5	R	515
	049	4	101.2	−82.6	1.9	R	580

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Table 2–*continued*

Unit	Code	Samples	D (° ETN)	I (° down)	α_{95}	Polarity	T_C (°C)
Rhum coastal dykes							
	A03T053	2	252.7	−79.7	6.2	R	570
	055	4	156.3	−73.7	3.3	R	570
	056	4	254.4	−74.0	10.9	R	575
	061	4	161.0	−67.6	3.1	R	560
	063	4	161.1	−69.3	4.3	R	475–600
	064	4	172.0	−58.6	2.0	R	530
	104	4	159.1	−75.1	1.7	R	530
	110	4	131.5	−65.2	2.0	R	520
	110A	3	192.6	−74.9	12.9	R	550
	116	4	192.6	−57.2	3.5	R	560
	120	4	150.1	−74.9	1.6	R	550
	125	4	175.0	−72.0	1.0	R	550
	129	4	179.9	−68.8	1.1	R	540
	133	4	174.5	−60.5	0.7	R	560
	134	4	159.6	−76.4	2.0	R	580
	136	2	163.7	−59.7	1.3	R	540
	137	4	163.8	−55.0	3.4	R	565
	143A	4	195.3	−19.2	3.1	I	520
	161	4	190.4	−59.6	2.2	R	555
	161S	4	176.3	−59.0	1.2	R	
	162	3	179.0	−64.3	1.4	R	550
	171	4	161.4	−69.3	1.4	R	555
	172	4	196.3	−69.6	1.1	R	545
	173	4	175.4	−58.9	1.5	R	550
	175	4	179.2	−73.9	4.7	R	550
	179	4	182.2	−59.5	0.7	R	450
	182	4	212.0	−62.5	4.0	R	520
	187	3	122.1	−67.4	6.9	R	530
	194	4	180.5	−69.0	2.5	R	470
	198	4	No mean disparate directions				480
	199	4	207.9	−70.0	0.5	R	450
	203	4	185.4	−65.2	1.1	R	450
	204	4	225.2	−57.2	6.3	R	470
	205	4	187.6	−68.7	3.9	R	480
	207	4	168.1	−62.7	1.3	R	450
	208	4	167.6	−64.9	1.8	R	450
	209	4	165.8	−47.4	1.4	R	460
	212	4	226.6	−55.6	3.9	R	505
	218	4	107.0	−69.7	10.2	R	490
	228	4	172.6	−64.0	2.8	R	480
	228A	4	192.2	−37.1	11.0	R	550
	229	4	177.3	−60.5	2.2	R	560
	231	4	153.7	−61.3	1.5	R	540
	231A	4	201.3	−64.0	1.4	R	550
	232	4	181.8	−57.9	1.3	R	500
	233	4	198.5	−54.4	5.2	R	565
	234	3	183.8	−52.0	6.9	R	570–600
	235	4	177.1	−70.2	3.9	R	560
	240	3	201.7	−67.2	1.8	R	560
	241	4	181.8	−62.9	3.5	R	570
	242	4	301.8	−82.4	14.8	R	600
	243	4	224.2	−81.5	1.0	R	560

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Table 2–*continued*

Unit	Code	Samples	D (° ETN)	I (° down)	α_{95}	Polarity	T_C (°C)
Rhum coastal dykes							
	A03T244	4	177.6	−64.3	2.2	R	565
	245	4	184.9	−66.0	2.2	R	560
	246	3	188.1	−47.2	4.7	R	580
	247	3	253.7	−75.7	1.3	R	550
	248	4	199.3	−74.9	2.5	R	575
(Overall mean of reversed units) (excluding 242)		57 (units)	179.8	−67.2	3.1	R	
Canna lavas							
Lowest	A03T520	4	182.4	−70.1	4.6	R	300 (550)
	521	4	187.8	−65.4	1.4	R	300 (500)
	522	4	116.4	−71.8	1.0	R	280 (530)
	523	4	152.2	−62.2	1.1	R	350 (560)
	524	4	168.3	−59.2	1.8	R	250 (520)
	525	4	157.1	−67.3	1.8	R	280 (560)
	526	5	167.6	−65.8	4.2	R	250 (530)
	527	4	128.2	−51.9	6.1	R	300 (520)
	528	4	188.6	−51.6	1.5	R	350–500
	529	4	168.3	−69.4	2.0	R	290 (500)
Highest	530	4	205.0	−51.6	2.4	R	270 (480)
(Overall mean) (excluding 527)		10 (units)	173.2	−65.2	7.5	R	
Canna dykes							
	A03T501	4	201.8	−22.8	1.4	I	220?(580)
	502			not collected			
	503			not collected			
	503A	4	341.3	82.9	2.8	N	340 (550)
	504	4	195.0	−68.3	3.5	R	300 (480)
	505	4	303.5	76.6	2.4	N	430 (580)
	506	4	217.8	−58.0	1.4	R	280 (560)
	507	4	208.7	−15.6	3.0	I	220 (550)
	508	4	205.0	−79.9	1.4	R	360 (520)
	509	4	217.5	−56.3	1.0	R	~450 (560)
	510			not collected			
	511	4	188.7	−49.4	1.2	R	330 (540)
	512	4	191.5	−49.6	0.9	R	~290 (550)
	513	4	168.3	−40.7	3.1	R	230 (520)
	514	4	211.6	−48.5	2.0	R	440 (600)
	515	4	206.4	−64.1	0.8	R	~250 (500)
	516	4	181.3	−53.7	3.1	R	450/600 (640)
(Overall means) (units)							
N		2	316.4	80.3	19.6	N	
R		10	196.5	−57.9	8.9	R	
N + R		12	199.8	−51.4	32.5	–	

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Results

POLARITY AND DIRECTIONS OF MAGNETIZATION

Twenty-five millimetres long samples cut from the cores were used for measurements of remanent magnetization; alternating field cleaning was used for four or more independently orientated samples for all units, and some additional samples from selected units were thermally cleaned. Unit mean directions have been computed using the method of Dagley & Ade-Hall (1970) and are given in Table 2 and Figs 2, 5, 6 and 7.

Rhum centre

Several samples had very strong NRM intensities (from $1.25–15.60 \times 10^{-2}$ A m^2 kg^{-1}) and directions different from their companions. This suggests magnetization by lightning and these samples have been discounted. The remaining samples give reversed directions of magnetization near 180°, −60° (Fig. 2). There is little evidence of overprinting either from af or thermal demagnetization studies (Fig. 3), and overall there were only small changes of direction during cleaning.

No trend or correlation with position can be identified in the layered ultrabasic body, and the directions have been combined as from a single unit (Table 2). The mean direction obtained is slightly steeper than that reported by Khan (1960) and is quite similar to the combined direction for granophyre and ultrabasic given by Oldershaw (1967) (Table 2).

The directions for the various components of the explosion breccia are not identical, but most cluster together (Fig. 2b) near the directions found for other units of the central complex (Fig. 2a).

Many of the blocks are very large (up to 90 m across) and samples were collected in three groups (310-1, 313-4, 315-20) over a wide area (~ 500 m). The differences between the gabbro blocks (310, 313, 315, 318, 319, 320) suggest that there has been little, if any, general remagnetization. If that is so, then the similarity of the directions of magnetization between dykes (311, 314) and their host blocks (310, 313), between the blocks and the nearby matrix (316, 317A) and between the breccia as a whole and the other units of the centre implies little change in field direction and little movement during this period of activity. On the other hand, the similarities in direction of magnetization of blocks and matrix (315/316, 318-20/317A) might argue for at least partial remagnetization, in which case the difference between the two sets would have to be accounted for by *later* tectonic movement.

The results from the Harris brecciated cumulate (305) suggest that both the blocks (~ 0.2 m) and the matrix within the small area sampled (~ 3 m) were magnetized at the same time. Twelve of the 14 cores gave directions of magnetization which cluster together close to the directions found for other units of the complex. The agreement of af and thermal cleaning for one of the other two cores suggests that its direction of magnetization is not the result of inadequate cleaning; one possible explanation is misorientation of the core.

Rhum coastal dykes

Of the 62 bodies (dykes and a sill) sample, 58 were reversely magnetized, two (042, 143A) intermediate (on the basis of a polar colatitude between 40° and 140° in the present geographical coordinate system; Wilson, Dagley & McCormack 1972b) and two (033, 198) uncertain because of the large scatter between cores. Fig. 4 is a histogram of α_{95} for all units; for the majority $\alpha_{95} < 7°$.

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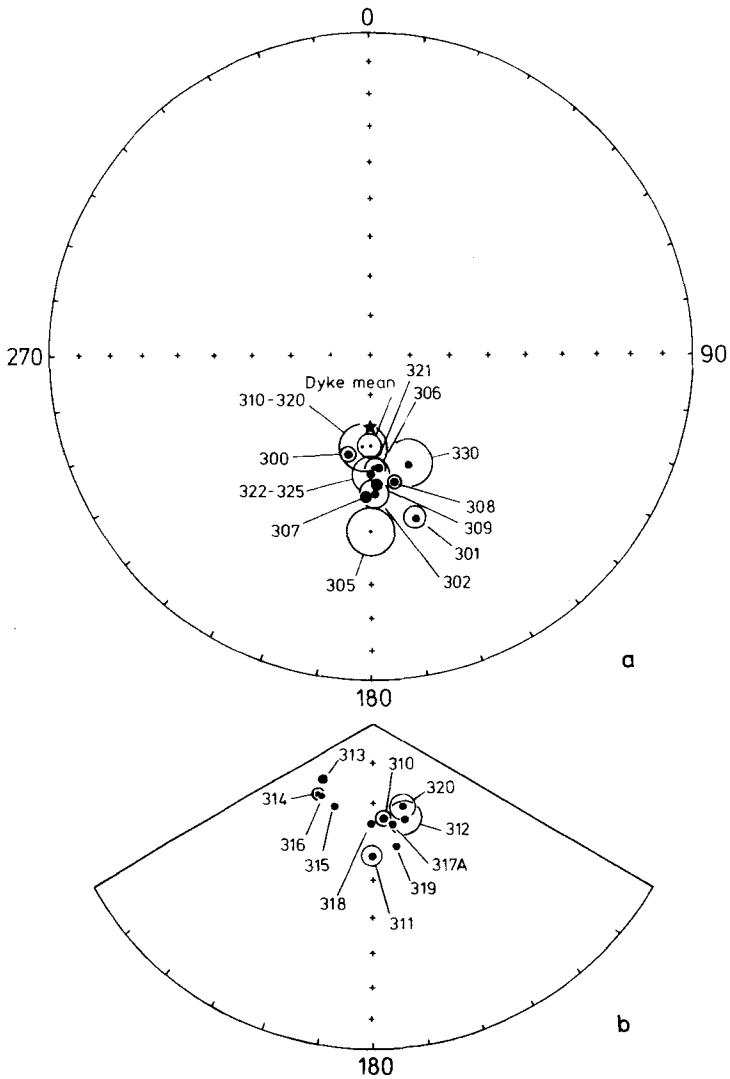


Figure 2. (a) Unit mean directions of magnetization for the Rhum centre (equal area projection), full circles – negative inclinations; crosses – positive inclinations; ★ direction of centred axial dipole field. (b) Individual directions for explosion breccia.

Unit mean directions are shown in Fig. 5. The overall mean of the 57 reversed dykes with well-defined directions (i.e. excluding 242) is 179.8°, −67.2°, with an α_{95} of 3.6°. This is shallower than the dip of a centred axial dipole, which is −72° at this latitude. The reversely magnetized dykes of Skye (Wilson *et al.* 1974), Mull (Ade-Hall *et al.* 1972), Arran (Dagley *et al.* 1978) and Lundy (Mussett, Dagley & Eckford 1976) also give mean directions shallower than the dipole field.

We find no correlation of direction of magnetization with dyke trend.

Although it is possible that we have missed any normal dykes by our semi-random sampling, because they occur in localized groups, it seems more likely that there are few, if any, normal dykes in the coastal group, unlike other areas in the province (see Discussion).

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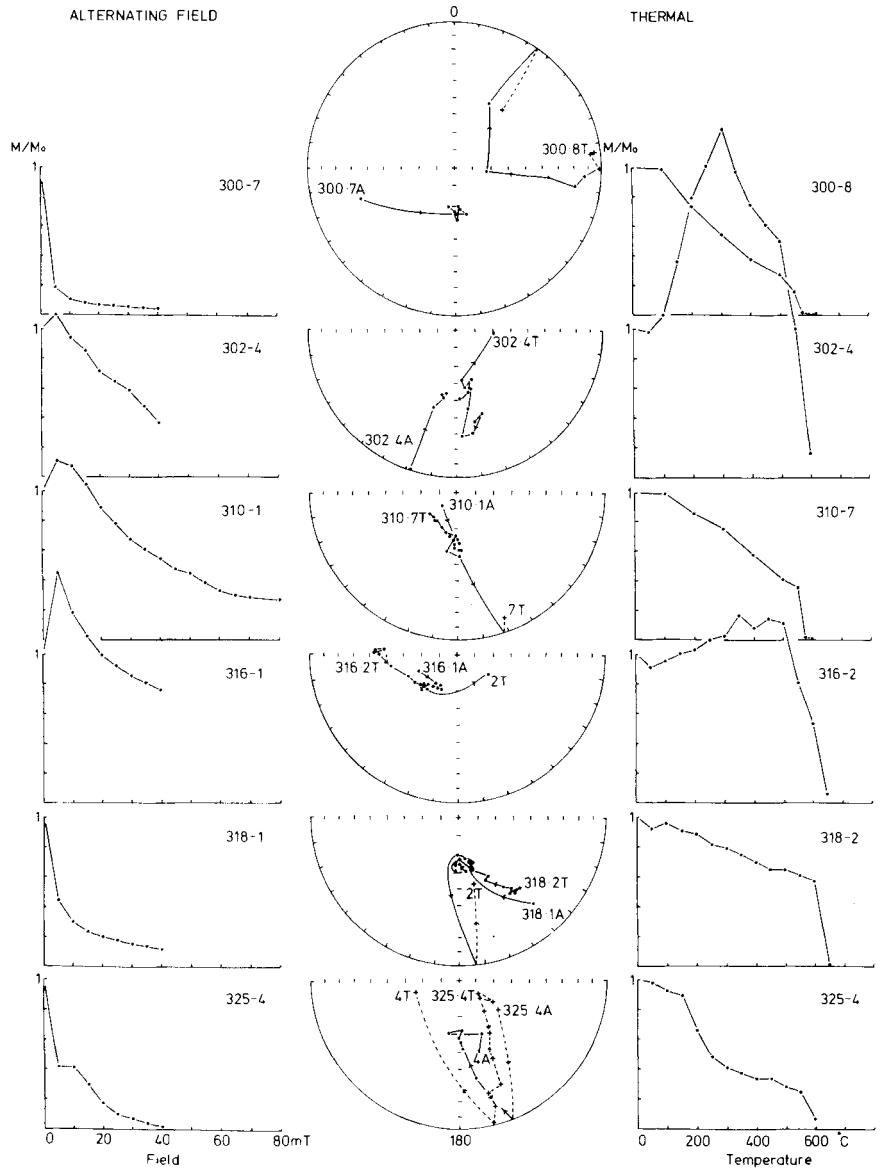


Figure 3. Examples of alternating field and thermal demagnetization of samples from the Rhum centre. M/M_0 = normalized intensity of magnetization. Symbols on equal area plots as in Fig. 2.

Rhum lavas and dykes cutting them

The directions of magnetization of all the 15 lavas sampled have negative inclinations: 12 are reversely magnetized, two have intermediate directions and for the remaining one the cores yield disparate directions. Of the dykes which cut the lavas, one (402D) is reversely magnetized and the other three are normally magnetized so, unlike the coastal dykes, the majority are normally magnetized.

The unit mean directions (Fig. 6), although individually well defined, show a considerable spread, including some shallower and some steeper than the field of a centred axial dipole.

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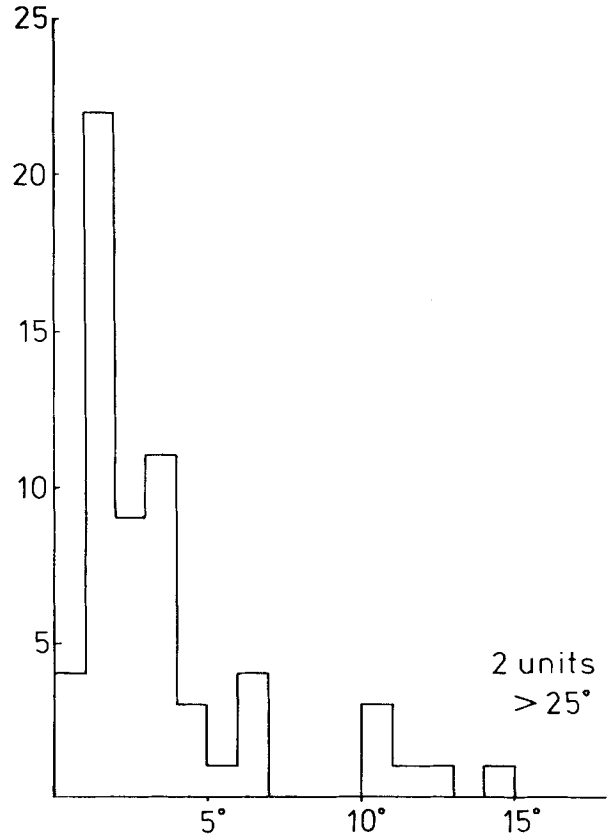


Figure 4. Histogram of α_{95} for Rhum coastal dykes.

The only subgroup separately distinguishable is that consisting of the Orval basalts, whose members have more easterly directions.

Canna dykes

Most of the dykes are reversely magnetized and the majority of these have directions of magnetization shallower than the field of a centred axial dipole (Fig. 7). Dykes 501 and 507 have very shallow inclinations and have been classed as having intermediate polarity (Wilson, Dagley & Ade-Hall 1972a). Both are very much wider dykes than the rest and very nearly colinear, so that they may be the same body. The two normally magnetized units 503A and 505 have directions of magnetization steeper than a centred axial dipole field.

Samples from odd numbered bodies (including 503A) were thermally cleaned. Although there was fairly good agreement between the af and thermally cleaned results, the directions obtained by the thermal technique did not converge into such a tight grouping as those from the af technique, as has been found elsewhere in the British Tertiary igneous province.

Canna lavas

All the lavas are reversely magnetized with, overall, shallower inclinations than would occur with a central axial dipole field. The unit mean directions (Fig. 7) are individually well

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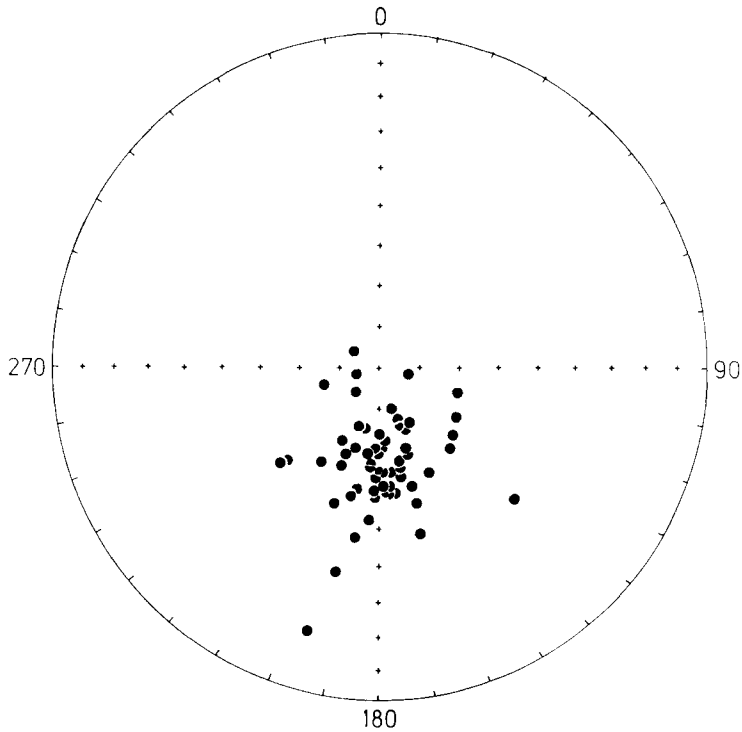


Figure 5. Unit mean directions of magnetization for the Rhum coastal dykes. Symbols as for Fig. 2.

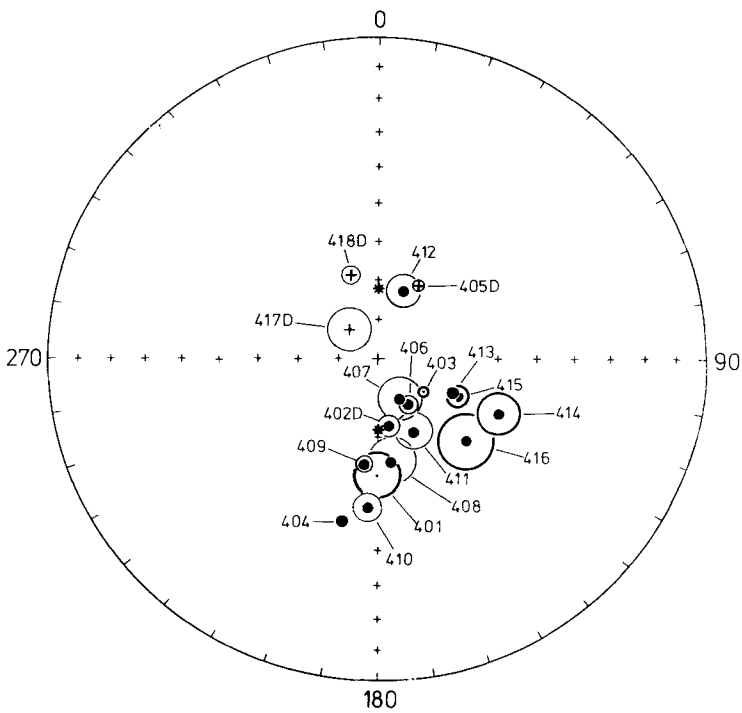


Figure 6. Unit mean directions of magnetization for the Rhum lavas and associated dykes. Symbols as for Fig. 2.

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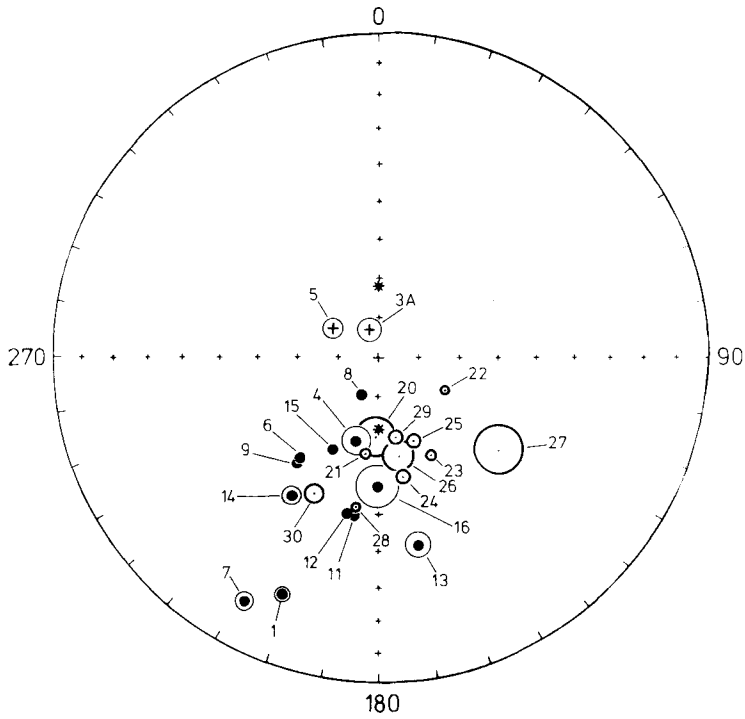


Figure 7. Unit mean directions of magnetization for the Canna lavas and associated dykes. Symbols as for Fig. 2.

defined and group quite closely together, with the exception of 522 and 527, which have rather easterly declinations, and 528 and 530 which are shallower and more westerly than the rest.

For the even numbered units additional samples were thermally cleaned. The agreement between thermal and af cleaning for the same core was not in general very good, but neither was it close between cores from the same unit thermally cleaned. This is in contrast to the tight grouping found between the samples cleaned in alternating fields. In the cases which differed, the directions of magnetization after each step of thermal cleaning did not even follow a systematic trend, demonstrating the instability of these rocks to thermal treatment.

THERMOMAGNETIC ANALYSIS

Small pieces from the cores were used for thermomagnetic measurements in a strong field to determine the Curie points: these are listed in Table 2.

Rhum centre

Heating curves showed that all bodies had single high temperature Curie points in the range 570° to 600°C, consistent with deuteric alteration (Ade-Hall, Palmer & Hubbard 1971). There are no differences which can be related to rock type.

Rhum coastal dykes

The distribution of Curie points shown in Fig. 8 is bimodal. The principal group lies between 515° and 600°C, the smaller group between 450° and 505°C. The thermomagnetic curves for

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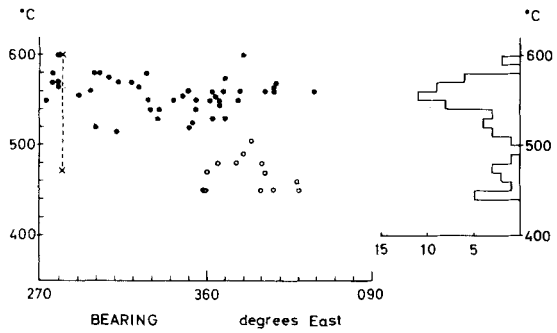


Figure 8. Curie points of Rhum coastal dykes as function of dyke trend.

the main group are simple: on cooling the samples indicate the same or a slightly lower value of T_C and a lower magnetization. The second group generally cool from a higher T_C but give a lower magnetization so that heating and cooling curves cross.

Discounting one case of a double Curie point, all those lower than 510°C belong to dykes trending between 358° and 050°, although there are dykes with trends in this range with high Curie points.

These low Curie point, north to north-east trending dykes (179–228) were sampled between Kilmory Bay and the mouth of Loch Scresort. Dykes of this trend sampled elsewhere (e.g. the north shore of Loch Scresort) all have high Curie points. Thus there appears to be a separate group of dykes with trends to east of north *and* with low Curie points, although their directions of magnetization are not distinct.

Speight (1972) has suggested that the Rhum coastal dykes are amongst the earliest in the Small Isles and that in turn the Small Isles dykes may be older than those of the Skye Swarm. In Skye, he suggests that the dykes of the Broadford–Applecross subswarm are earlier than the main north-west–south-east swarms, so that they could be contemporary with those in the Small Isles. We note that the north-east trending dykes of Rhum lie close to the line of the Broadford–Applecross subswarm, so the dykes with low Curie points may have been intruded earlier in the succession than those with high Curie points.

The fact that these Curie points are lower suggests – but does not prove – that the similarity of mean directions for the dykes and for the centre is not the result of remagnetization by the centre but is due to similarity of the directions of the magnetizing fields throughout the intrusive activity.

Rhum lavas

Most of the flows display smooth geniculate strong-field heating curves falling to a low temperature Curie point (200°–300°C). Of these, many, on cooling from temperatures above this Curie point, become magnetized at higher temperatures (400°–500°C), indicating a change during heating. These results suggest that the lavas cannot have been oxidized by thermal alteration (e.g. by burial) to any significant extent since intrusion.

The cooling Curie points distinguish the olivine basalts (400, 401, 403) with a mean of 440 ± 15 from the basaltic–andesites (406–409) with a mean of 484° ± 22°C. The latter also have a higher magnetization on cooling, in contrast to the remainder which have a lower magnetization, so that the cooling curve crosses the heating curve.

The Orval (B′) basalts (413–416) generally have a higher Curie point on heating and a lower one on cooling (379° ± 25°C) than the Fionchra (B) basalts (440° ± 15°C).

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Canna dykes

The thermomagnetic curves indicate a range of Curie points between 200° and 600°C; A03T516 appeared to have two at 450° and 600°C. All samples changed during heating to give Curie points of 480° to 600°C on cooling. In general the change of slope marking the Curie point was fairly sharp, the exceptions being 501 and 507 for which the curves were concave downwards with a steady decrease in magnetization but suggesting a Curie point around 220°C.

Canna lavas

All the rapid heating curves gave clear-cut low-temperature Curie points in the range 250°–300°C. Changes occurred during heating so the cooling curves were markedly different from the heating curve, with Curie points between 500°–560°C (mean 517° ± 33°C). The low Curie points suggest that the lavas have not been subjected to any significant deuteric or hydrothermal alteration; they also explain the instability of the rocks to thermal cleaning. It should be noted that these cooling Curie points are generally higher than those of the Rhum lavas; they are significantly different to those for the olivine basalts (Fionchra) and the hawaiites (Orval basalts) but their distribution overlaps that of the basaltic andesites (early mugearites of Fionchra). These differences possibly reflect differences in composition of the magnetic minerals, implying a different origin or a different history.

Discussion

The most striking result from this work is the great predominance of reversely magnetized rocks: the only definitely normally magnetized rocks are three of the four dykes cutting the Rhum lavas and two of the 14 dykes cutting the Canna lavas. All the lavas are reversed, as elsewhere in the British Tertiary igneous province (BTip), (Skye, Wilson *et al.* 1972a; Mull, Hall, Wilson & Dagley 1977; Northern Ireland, Wilson 1970; Eigg and Muck, Dagley & Mussett 1979) (except for a pillow lava in the Mull caldera, which may have been overprinted, Mussett, Dagley & Skelhorn 1980), but also the whole of the centre is reversed. This contrasts with other areas of the province. For instance, both polarities are found in the centres of Mull (Mussett *et al.* 1980), Arran (Dagley & Hodgson 1979; Dagley *et al.* 1978) and Skye (Dagley & Mussett 1980). In addition, of the 62 coastal dykes examined from Rhum, 58 are clearly reversed and the remaining four have negative inclinations, so it seems that this swarm contains few, if any, normally magnetized dykes. Other dyke swarms examined elsewhere all contain a proportion of normally magnetized dykes: Skye (Vaternish), 19 per cent of 87 (Wilson *et al.* 1974); Skye (Sleat), 19 per cent of 409 (unpublished work of this laboratory); Mull, 27 per cent of 165 (Ade-Hall *et al.* 1972); Arran, 17 per cent of 435 (Dagley *et al.* 1978) and Lundy, 8 per cent of 66 (Mussett *et al.* 1976).

The question arises whether this predominance of reversed polarity is primary, or due to overprinting by the heat of the centre. It is highly unlikely that the coastal dykes, although they came early in the succession, are overprinted because they are up to 5 km from the centre. It is also clear that the magnetism of the lavas is primary, since they are later than the centre and are cut by normally magnetized dykes. Thus any overprinting is confined to the centre itself (overprinting here is taken to include a significant delay in magnetization owing to continued high temperature, as well as to distinct reheating). The data from the Harris breccia suggest it was magnetized at the time of welding but attempts to determine whether the explosion breccia has been remagnetized after formation were inconclusive. The simplest interpretation of the results is that the magnetization is primary and no significant thermal overprinting has occurred.

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Even if overprinting or delayed magnetization did occur the delay is not likely to have been long, because the centre is not a large one and there is no evidence for prolonged development; the cooling of the larger and much more complex Mull centre probably only took a fraction of a million years (Mussett *et al.* 1980). The duration of normal polarity intervals at about this time was about 0.5 Ma (La Breque, Kent & Cande 1977; Hailwood *et al.* 1979) so it is unlikely that such an interval, although short compared to the reversed intervals, could have escaped being recorded in some of the rocks if it had occurred.

Turning to the polarity magnetostratigraphy, the evidence therefore shows that the development of the Rhum Tertiary igneous rocks began in a reversed period, which probably continued throughout the emplacement of the centre. The subsequent lavas also formed in a reversed period, followed – probably only shortly afterwards – by the normally magnetized dykes. Thus the polarity changed near the end of activity. This indicates only a simple R–N sequence; however, an unknown period elapsed while the centre was eroded prior to the extrusion of the lavas upon it, so it is possible that one or more normal intervals could have occurred in this time, giving R–...N...R–N.

This uncertainty could most easily be resolved by radiometric dating. Conventional K-Ar dating (unpublished work of this laboratory) gave a spread from 35 to 102 Ma, although the majority were in the upper fifties, a range not unknown for conventional K-Ar dates for the BTip. Because of this, the $^{40}Ar/^{39}Ar$ step-heating method has been tried, and has yielded a plateau age of 58.7 ± 0.6 (1σ) Ma for the Western Granophyre (unpublished work of this laboratory). Other rocks tried so far have failed to yield plateaux.

It is likely that the R–N polarity sequence recorded by the Canna rocks is contemporaneous with that in Rhum, for interbedded conglomerates indicate (Allwright, private communication) that the Canna lavas – like the Rhum lavas – post-date the erosion of the Rhum centre. This evidence supports Speight's (1972) explanation of the dyke densities on the adjacent shores of Rhum and Canna, which he suggested could be due to the Canna lavas covering the earlier coastal dykes. The Rhum lavas also are cut by very few dykes.

If the Rhum and Canna lavas are contemporaneous (and perhaps came from the same source – Allwright, private communication), in spite of the somewhat different thermomagnetic behaviour, the normally magnetized dykes of the two areas are probably coeval. An examination of the directions of magnetization of the lavas, rather than just polarities, does not help because although the mean directions (Table 2) are different, the ovals of confidence overlap.

How do the polarity periods of Rhum and Canna relate to those of the rest of the British Tertiary igneous province? On the basis of the age of 58.6 ± 0.9 Ma quoted above for the Western Granophyre, the Rhum centre is broadly contemporaneous with much activity elsewhere in the province (see Mussett 1980, for a summary of the most reliable dates). It is also known that polarity sequences of at least R–N–R occurred during the development of Mull (Mussett *et al.* 1980), Arran (Dagley & Hodgson 1979) and Skye (Dagley & Mussett 1980). However, the precession of the available dates is insufficient to link these polarities firmly together. Tentatively, the reversed period of Rhum was the same as that in which the Mull lavas formed (58 ± 1 Ma), Sankey & Mussett 1979). Then the normal period during which the late dykes formed would coincide with that in which the latter parts of the Mull centre developed (Mussett *et al.* 1980) *provided* that the hiatus between the Rhum centre and lavas did not span one or more N–R pairs. The settling of this point must await reliable dates.

Conclusions

(1) All the rocks measured have reversed polarity, except for three dykes of four cutting the Rhum lavas, and two of 14 cutting the Canna lavas.

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(2) Overprinting or delayed cooling is unlikely to have obscured the existence of a normal period.

(3) The lavas of Canna and Rhum are probably contemporaneous, as are the normal dykes which cut them.

(4) The Tertiary rocks may have been formed in two polarity intervals R–N, about 59 Ma ago. However, one or more pairs of polarity inversions may have occurred in the hiatus between the formation of the Rhum centre and the lavas. If this is so, the Canna lavas and dykes, and the Rhum lavas and dykes, represent a different R period than the centre.

(5) A group of Rhum coastal dykes can be distinguished by having *both* trends east of north and low Curie points, although their directions of magnetization are not distinct.

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Mrs A. Wall and Mrs M. Catterall did most of the measuring and Mr A. G. McCormack carried out computer analyses.

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